

Definition of the Laplace transform

a special type of integral transform is called the Laplace transform

definition:

let f be a function defined for $t \geq 0$ then, the integral

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

is the Laplace transform of f , if the integral converges

$e^{-st} \Rightarrow$ kernel function

$f(t) \Rightarrow$ function of t

So all you are doing here is taking the function $f(t)$ you are given and multiplying it with e^{-st} and setting up the limit you are given in an integral

you will need these for part two of finding $\mathcal{L}\{f(t)\}$:

Theorem 7.1.1 Transforms of Some Basic Functions

- (a) $\mathcal{L}\{1\} = \frac{1}{s}$
- (b) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, n = 1, 2, 3, \dots$
- (c) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$
- (d) $\mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2}$
- (e) $\mathcal{L}\{\cos kt\} = \frac{s}{s^2 + k^2}$
- (f) $\mathcal{L}\{\sinh kt\} = \frac{k}{s^2 - k^2}$
- (g) $\mathcal{L}\{\cosh kt\} = \frac{s}{s^2 - k^2}$

use the Definition of the Laplace theorem to find $\mathcal{L}\{f(t)\}$

$$f(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 1, & t \geq 1 \end{cases}$$

$$\int_0^1 t e^{-st} dt + \int_1^{\infty} 1 \cdot e^{-st} dt$$

So all we did here was take the given $f(t)$ and multiplied it by e^{-st} and took the given limit and set it as an integral

$$uv - \int v du, \text{ by parts}$$

$$u = t$$

$$du = 1 dt$$

$$dv = e^{-st}$$

$$v = \int e^{-st} dt$$

$$v = -\frac{1}{s} e^{-st}$$

$$\left(-\frac{te^{-st}}{s} - \int -\frac{1}{s} e^{-st} \right) \Rightarrow \left(-\frac{te^{-st}}{s} - \frac{1}{s^2} e^{-st} \right)$$

$$\left[\left(-\frac{te^{-st}}{s} - \frac{1}{s^2} e^{-st} \right) \right]_0^1 \Rightarrow \left[\left(-\frac{e^{-s}}{s} - \frac{1}{s^2} e^{-s} \right) - \left(0 - \frac{1}{s^2} \right) \right]$$

$$\Rightarrow -\frac{e^{-s}}{s} - \frac{1}{s^2} e^{-s} + \frac{1}{s^2}$$

★ Remember we are applying $\int_a^b f(x) = [F(b) - F(a)]$

$$\int_1^{\infty} 1 \cdot e^{-st} dt$$

• this is an improper integral so you will need to take the limit $\lim_{m \rightarrow \infty} \int_1^m f(t) dt$ take integral then apply the limit

$$\lim_{m \rightarrow \infty} \int_1^m e^{-st} dt \Rightarrow \lim_{m \rightarrow \infty} \left[-\frac{1}{s} e^{-st} \right]_1^m$$

$$\lim_{m \rightarrow \infty} \left[\left(-\frac{1}{s} e^{-sm} \right) - \left(-\frac{1}{s} e^{-s} \right) \right]$$

$$\lim_{m \rightarrow \infty} \left[-\frac{1}{s} e^{-sm} + \frac{1}{s} e^{-s} \right] \Rightarrow \left[-\cancel{\frac{1}{s e^{-s \infty}}} + \frac{1}{s} e^{-s} \right]$$

$$\cancel{\frac{-e^s}{s}} - \frac{1}{s^2} e^{-s} + \frac{1}{s^2} + \cancel{\frac{1}{s} e^s} \Rightarrow -\frac{1}{s^2} e^{-s} + \frac{1}{s^2} = \boxed{\frac{1}{s^2} (1 - e^{-s})}$$

ex #2:

let $f(t) = e^{-2t-s}$, use the definition to find $\mathcal{L}\{f(t)\}$:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\int_0^{\infty} e^{-st} (e^{-2t-s}) dt \Rightarrow \lim_{b \rightarrow \infty} \int_0^b e^{-st-2t-s} dt$$

$$\lim_{b \rightarrow \infty} \int_0^b e^{-st-2t} \cdot e^{-s} dt \rightarrow \lim_{b \rightarrow \infty} \int_0^b e^{-t(s+2)} \cdot e^{-s} dt$$

$$e^{-s} \lim_{b \rightarrow \infty} \int_0^b e^{-t(2+s)} dt \Rightarrow e^{-s} \lim_{b \rightarrow \infty} \left[\frac{1}{(2+s)} e^{-t(2+s)} \right]_0^b$$

$$\lim_{b \rightarrow \infty} \left[\left(\frac{1}{(2+s)} e^{-b(2+s)} \right) - \left(\frac{1}{(2+s)} \right) \right]$$

$$e^{-s} \left[\left(\frac{1}{(2+s)} e^{-\infty(2+s)} \right) - \left(\frac{1}{(2+s)} \right) \right]$$

$$= \boxed{\frac{-e^{-s}}{(2+s)}}$$

ex #3:

use the definition of laplace theorem to find $\mathcal{L}\{f(t)\}$:

$$f(t) = \begin{cases} -1, & 0 \leq t \leq 1 \\ 1, & t \geq 1 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) \cdot dt$$

$$\int_0^1 -e^{-st} dt + \int_1^{\infty} e^{-st} dt$$

$$\left[\frac{1}{s} e^{-st} \right]_0^1 + \lim_{b \rightarrow \infty} \int_1^b e^{-st} dt$$

$$\left[\frac{1}{s} e^{-st} \right]_0^1 + \lim_{b \rightarrow \infty} \left[-\frac{1}{s} e^{-st} \right]_1^b$$

$$\left[\left(\frac{1}{s} e^{-s} \right) - \left(\frac{1}{s} \right) \right] + \lim_{b \rightarrow \infty} \left[\left(-\frac{1}{s} e^{-sb} \right) - \left(-\frac{1}{s} e^{-s} \right) \right]$$

$$\frac{1}{s} e^{-s} - \frac{1}{s} + \left[\cancel{\frac{-1}{s} e^{-\infty s}}^0 + \frac{1}{s} e^{-s} \right]$$

$$\frac{1}{s} e^{-s} - \frac{1}{s} + \frac{1}{s} e^{-s}$$

$$\boxed{\frac{2}{s} e^{-s} - \frac{1}{s}}$$

Find the $\mathcal{L}\{f(t)\}$ using the definition:

$$f(t) = \begin{cases} 4, & 0 \leq t \leq 2 \\ 0, & t \geq 2 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\int_0^2 4e^{-st} dt + \int_2^{\infty} 0 \cdot e^{-st} dt$$

$$\left[\frac{-4}{s} e^{-st} \right]_0^2 + [0]$$

← the reason why this is zero instead of "c" is because this is a definite integral you don't add "c"

$$\left[\left(-\frac{4}{s} e^{-2s} \right) - \left(-\frac{4}{s} \right) \right]$$

$$= \boxed{-\frac{4}{s} e^{-2s} + \frac{4}{s}}$$

Find the $\mathcal{L}\{f(t)\}$ using the definition:

$$f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\int_0^1 t e^{-st} dt + \int_1^{\infty} e^{-st} dt$$

uv- $\int v du$, $\lim_{b \rightarrow \infty}$

$$u = t$$

$$du = dt$$

$$dv = e^{-st}$$

$$v = \int e^{-st}$$

$$v = -\frac{1}{s} e^{-st}$$

$$\lim_{b \rightarrow \infty} \int_1^b e^{-st} dt$$

$$\lim_{b \rightarrow \infty} \left[-\frac{1}{s} e^{-st} \right]_1^b$$

$$-\frac{t}{s} e^{-st} - \int -\frac{1}{s} e^{-st}$$

$$-\frac{t}{s} e^{-st} + \frac{1}{s} \left[-\frac{1}{s} e^{-st} \right]$$

$$\left[-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^1$$

$$\left[-\frac{1}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right]_0^1 + \lim_{b \rightarrow \infty} \left[-\frac{1}{s} e^{-st} \right]_1^b$$

$$\left[\left(-\frac{1}{s} e^{-s} - \frac{1}{s^2} e^{-s} \right) - \left(0 - \frac{1}{s^2} \right) \right] + \lim_{b \rightarrow \infty} \left[\left(-\frac{1}{s} e^{-bs} \right) - \left(-\frac{1}{s} e^{-s} \right) \right]$$

$$-\frac{1}{s} e^{-s} - \frac{1}{s^2} e^{-s} + \frac{1}{s^2} + \left[-\cancel{\frac{1}{s} e^{-\infty s}} + \frac{1}{s} e^{-s} \right]$$

$$\cancel{-\frac{1}{s} e^{-s}} - \frac{1}{s^2} e^{-s} + \frac{1}{s^2} + \cancel{\frac{1}{s} e^{-s}}$$

$$= \boxed{-\frac{1}{s^2} e^{-s} + \frac{1}{s^2}}$$

Find the $\mathcal{L}\{f(t)\}$ using the definition:

$$f(t) = \begin{cases} 2t+1, & 0 \leq t < 1 \\ 0, & t \geq 1 \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\int_0^1 e^{-st} (2t+1) dt + \int_1^{\infty} 0 \cdot e^{-st} dt$$

$$uv - \int v du, \text{ let } + [0]$$

$$u = 2t+1$$

$$du = 2 dt$$

$$dv = e^{-st}$$

$$v = \int e^{-st} dt$$

$$v = -\frac{1}{s} e^{-st}$$

$$(2t+1) \left(-\frac{1}{s} e^{-st}\right) - \int -\frac{1}{s} e^{-st} \cdot (2 dt)$$

$$-\frac{2t}{s} e^{-st} - \frac{1}{s} e^{-st} + \frac{2}{s} \int e^{-st} dt$$

$$-\frac{2t}{s} e^{-st} - \frac{1}{s} e^{-st} + \frac{2}{s} \cdot \left[-\frac{1}{s} e^{-st}\right]$$

$$\left[-\frac{2t}{s} e^{-st} - \frac{1}{s} e^{-st} - \frac{2}{s^2} e^{-st}\right]_0^1$$

$$\left[\left(-\frac{2}{s} e^{-s} - \frac{1}{s} e^{-s} - \frac{2}{s^2} e^{-s}\right) - \left(0 - \frac{1}{s} - \frac{2}{s^2}\right)\right]$$

$$\boxed{-\frac{2}{s} e^{-s} - \frac{1}{s} e^{-s} - \frac{2}{s^2} e^{-s} + \frac{1}{s} + \frac{2}{s^2}}$$

Find $\mathcal{L}\{f(t)\}$ using the definition:

$$f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ 0, & t \geq \pi \end{cases}$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) \cdot dt$$

$$\int_0^{\pi} e^{-st} \cdot \sin t \, dt + \int_{\pi}^{\infty} 0 \cdot e^{-st} \, dt$$

$$\int e^{at} \sin(bt) = \frac{e^{at} (a \sin bt - b \cos bt)}{a^2 + b^2}$$

$$\left[\frac{e^{-st} (-s \sin t - \cos t)}{s^2 + 1} \right]_0^{\pi}$$

• $b=1$ because if you look at the equation you want what is next to "t"

$$\left[\left(\frac{e^{-\pi s} (-s(0) - (-1))}{s^2 + 1} \right) - \left(\frac{(1)(0) - 1}{s^2 + 1} \right) \right]$$

$$\frac{e^{-\pi s} (+1)}{s^2 + 1} + \frac{1}{s^2 + 1}$$

$$= \boxed{\frac{e^{-\pi s} + 1}{s^2 + 1}}$$

For this next part all we are doing is using the 7.1.1 theorem to plug into the equation to solve for $\mathcal{L}\{f(t)\}$

ex #1

let $f(t) = t^2 - e^{-9t} + 5$ Find $\mathcal{L}\{f(t)\}$:

$$\mathcal{L}\{t^2\} - \mathcal{L}\{e^{-9t}\} + \mathcal{L}\{5\}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{n!}{s^{n+1}} & \frac{1}{s-k} & \frac{1}{s} \end{array}$$

$$\boxed{\frac{2!}{s^3} - \frac{1}{s+9} + \frac{5}{s} = f(s)}$$

ex #2

let $f(t) = \cos(5t) + \sin(4t)$, Find $\mathcal{L}\{f(t)\}$:

$$\mathcal{L}\{\cos(5t)\} + \mathcal{L}\{\sin(4t)\}$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \frac{s}{s^2+k^2} & & \frac{k}{s^2+k^2} \end{array}$$

$$\boxed{\frac{s}{s^2+25} + \frac{4}{s^2+16}}$$

use theorem 7.1.1 to find $\mathcal{L}\{f(t)\}$

$$f(t) = 2t^4$$

$$2 \mathcal{L}\{t^4\} \quad \bullet \quad t^n = \frac{n!}{s^{n+1}}$$

$$2 \left(\frac{4!}{s^5} \right) = \boxed{\frac{48}{s^5}}$$

use theorem 7.1.1 to find $\mathcal{L}\{f(t)\}$:

$$f(t) = t^5 \quad \bullet \quad t^n = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}\{t^5\} \Rightarrow \boxed{\frac{5!}{s^6}}$$

use the theorem 7.1.1 to find $\mathcal{L}\{f(t)\}$:

$$f(t) = 4t - 10 \Rightarrow 4 \mathcal{L}\{t\} - \mathcal{L}\{10\}$$

$$\boxed{\frac{4}{s^2} - \frac{10}{s}}$$

$$\bullet \quad t^n = \frac{n!}{s^{n+1}} \quad \bullet \quad 1 = \frac{1}{s}$$

use the theorem from 7.1.1 to find $\mathcal{L}\{f(t)\}$:

$$f(t) = 7t + 3 \rightarrow \mathcal{L}\{t\} + \mathcal{L}\{3\}$$
$$\bullet t^n = \frac{n!}{s^{n+1}} \quad \bullet 1 = \frac{1}{s}$$

$$= \boxed{\frac{7}{s^2} + \frac{3}{s}}$$

using the theorem from 7.1.1 Find $\mathcal{L}\{f(t)\}$

$$f(t) = t^2 + 6t - 3 \rightarrow \mathcal{L}\{t^2\} + 6\mathcal{L}\{t\} - \mathcal{L}\{3\}$$
$$\bullet t^n = \frac{n!}{s^{n+1}} \quad \bullet t^n = \frac{n!}{s^{n+1}} \quad \bullet 1 = \frac{1}{s}$$

$$= \boxed{\frac{2!}{s^3} + \frac{6}{s^2} - \frac{3}{s}}$$

using the theorem 7.1.1 Find $\mathcal{L}\{f(t)\}$:

$$f(t) = -4t^2 + 16t + 9 \rightarrow -4\mathcal{L}\{t^2\} + 16\mathcal{L}\{t\} + \mathcal{L}\{9\}$$
$$\bullet t^n = \frac{n!}{s^{n+1}} \quad \bullet t^n = \frac{n!}{s^{n+1}} \quad \bullet 1 = \frac{1}{s}$$

$$-4\left(\frac{2!}{s^3}\right) + 16\left(\frac{1}{s^2}\right) + 9\left(\frac{1}{s}\right)$$

$$\boxed{-\frac{8}{s^3} + \frac{16}{s^2} + \frac{9}{s}}$$

using the theorem 7.1.1 Find $\int \{f(t)\} :$

$$f(t) = (t+1)^3$$

$$(t+1) \quad (t+1) \quad (t+1)$$

$$t^2 + t + t + 1$$

$$(t+1) \quad (t^2 + 2t + 1)$$

$$t^3 + \cancel{2t^2} + t + \cancel{t^2} + 2t + 1$$

$$t^3 + 3t^2 + 3t + 1$$

$$\int \{t^3\} + 3 \int \{t^2\} + 3 \int \{t\} + \int \{1\}$$

$$\frac{3!}{s^4} + 3 \left(\frac{2!}{s^3} \right) + 3 \left(\frac{1!}{s^2} \right) + \left(\frac{1!}{s} \right)$$

$$\boxed{\frac{6}{s^4} + \frac{6}{s^3} + \frac{3}{s^2} + \frac{1}{s}}$$

use theorem 7.1.1 to find $\int \{f(t)\} :$

$$f(t) = (2t-3)^3$$

$\Rightarrow$

$$(2t-3) \quad (2t-3) \quad (2t-3)$$

$$(2t-3) [4t^2 - 6t - 6t + 9]$$

$$(2t-3) \quad (4t^2 - 12t + 9)$$

$$8t^3 - 24t^2 + 18t - 12t^2 + 36t - 27$$

$$8t^3 - 36t^2 + 54t - 27$$

$$8t^3 - 36t^2 + 54t - 27$$

$$8 \int \{t^3\} - 36 \int \{t^2\} + 54 \int \{t\} - \int \{27\}$$

$$8 \left(\frac{3!}{s^4} \right) - 36 \left(\frac{2!}{s^3} \right) + 54 \left(\frac{1!}{s^2} \right) - \frac{27}{s}$$

$$\boxed{\frac{48}{s^4} - \frac{72}{s^3} + \frac{54}{s^2} - \frac{27}{s}}$$

use the theorem 7.1.1 to solve for $\mathcal{L}\{f(t)\}$

$$f(t) = 1 + e^{4t}$$

$$\bullet 1 = \mathcal{L}\left\{\frac{1}{s}\right\} \quad \bullet e^{kt} = \mathcal{L}\left\{\frac{1}{s-k}\right\}$$

$$= \boxed{\frac{1}{s} + \frac{1}{s-4}}$$

use theorem 7.1.1 to solve for $\mathcal{L}\{f(t)\}$

$$f(t) = t^2 - e^{-9t} + 5$$

$$\bullet t^n = \mathcal{L}\left\{\frac{n!}{s^{n+1}}\right\} \quad \bullet e^{kt} = \mathcal{L}\left\{\frac{1}{s-k}\right\} \quad \bullet 1 = \mathcal{L}\left\{\frac{1}{s}\right\}$$

$$= \boxed{\frac{2!}{s^3} - \frac{1}{s+9} + \frac{5}{s}}$$

use theorem 7.1.1 to find $\mathcal{L}\{f(t)\}$

$$f(t) = 4t^2 - 5\sin(3t)$$

$$\bullet t^n = \frac{n!}{s^{n+1}} \quad \bullet \sin kt = \frac{k}{s^2 + k^2}$$

$$4\left(\frac{2!}{s^3}\right) - 5\left(\frac{3}{s^2 + 9}\right)$$

$$= \boxed{\frac{8}{s^3} - \frac{15}{s^2 + 9}}$$

use theorem 7.1.1 to find $\mathcal{F}\{f(t)\}$:

$$f(t) = \cos(5t) + \sin(2t)$$

$$\bullet \cos(kt) = \frac{s}{s^2 + k^2}, \quad \bullet \sin(kt) = \frac{k}{s^2 + k^2}$$

$$= \boxed{\frac{s}{s^2 + 25} + \frac{2}{s^2 + 4}}$$